Application of Matrix and Determinant

Application of the System of Linear Equations

Problem-01:

Three types of food contain different amounts of vitamins A, B, and C per unit as follows:

Food	**Vitamin A**	**Vitamin B**	**Vitamin C**
Food I	3 units	9 units	12 units
Food II	6 units	9 units	15 units
Food III	9 units	3 units	9 units

A person requires **42 units of vitamin A**, **36 units of vitamin B**, and **69 units of vitamin C.** Determine how much of each food the person should consume to meet exactly these vitamin requirements using the **matrix method**.

Solution Using Matrix Method:

Step 1: Let

- x = amount of Food I
- y = amount of Food II
- z = amount of Food III

Set up the system of equations based on vitamin contributions:

$$\begin{aligned} 3x + 6y + 9z &= 42 \quad (\text{Vitamin A}) \\ 9x + 9y + 3z &= 36 \quad (\text{Vitamin B}) \\ 12x + 15y + 9z &= 69 \quad (\text{Vitamin C}) \end{aligned}$$

Step 2: Write in Matrix Form $AX = B$

$$A = \begin{bmatrix} 3 & 6 & 9 \\ 9 & 9 & 3 \\ 12 & 15 & 9 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 42 \\ 36 \\ 69 \end{bmatrix}$$

Step 3: Solve $X = A^{-1}B$

Final Answer:

The amounts of each food required are:

$$\boxed{x = 1, \quad y = 2, \quad z = 3}$$

That is:

- **1 unit of Food I**
- **2 units of Food II**
- **3 units of Food III**

Problem–02:

A, B, and C have Tk. 480, Tk. 760, and Tk. 710 respectively. They used these amounts to purchase three types of shares priced at x, y, and z respectively.

- A purchases 2 shares of price x, 5 shares of price y, and 3 shares of price z.
- B purchases 4 shares of price x, 3 shares of price y, and 6 shares of price z.
- C purchases 1 share of price x, 4 shares of price y, and 10 shares of price z.

Determine the values of x, y, and z.

Solution:

From the given data, we form the following system of equations:

$$2x + 5y + 3z = 480$$
$$4x + 3y + 6z = 760$$
$$x + 4y + 10z = 710$$

This can be written in matrix form as:

$$\begin{bmatrix} 2 & 5 & 3 \\ 4 & 3 & 6 \\ 1 & 4 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 480 \\ 760 \\ 710 \end{bmatrix}$$

Let A be the coefficient matrix, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, and B the constant matrix:

$$AX = B \Rightarrow X = A^{-1}B$$

Step 1: Compute Determinant of A

$$|A| = \begin{vmatrix} 2 & 5 & 3 \\ 4 & 3 & 6 \\ 1 & 4 & 10 \end{vmatrix} = -119$$

Step 2: Find Adjoint of A

$$\mathrm{Adj}(A) = \begin{bmatrix} 6 & -38 & 21 \\ -34 & 17 & 0 \\ 13 & -3 & -14 \end{bmatrix}$$

Step 3: Compute Inverse of A

$$A^{-1} = \frac{\mathrm{Adj}(A)}{|A|} = \frac{1}{-119} \begin{bmatrix} 6 & -38 & 21 \\ -34 & 17 & 0 \\ 13 & -3 & -14 \end{bmatrix}$$

Step 4: Multiply A^{-1} with B

$$X = A^{-1}B = -\frac{1}{119}\begin{bmatrix} 6 & -38 & 21 \\ -34 & 17 & 0 \\ 13 & -3 & -14 \end{bmatrix}\begin{bmatrix} 480 \\ 760 \\ 710 \end{bmatrix}$$

$$= -\frac{1}{119}\begin{bmatrix} 6\times 480 - 38\times 760 + 21\times 710 \\ -34\times 480 + 17\times 760 + 0\times 710 \\ 13\times 480 - 3\times 760 - 14\times 710 \end{bmatrix} = -\frac{1}{119}\begin{bmatrix} -11090 \\ -3400 \\ -5980 \end{bmatrix}$$

Step 5: Final Result

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{11090}{119} \\ \frac{3400}{119} \\ \frac{5980}{119} \end{bmatrix} \Rightarrow \boxed{x = \frac{11090}{119}, \quad y = \frac{3400}{119}, \quad z = \frac{5980}{119}}$$

Problem–03:

To control a certain crop disease, it is necessary to use:

- 8 units of **chemical A**
- 14 units of **chemical B**
- 13 units of **chemical C**

Three types of sprays are available:

- **Spray P** (per barrel) contains:
  1 unit of A, 2 units of B, and 3 units of C
- **Spray Q** (per barrel) contains:
  2 units of A, 3 units of B, and 2 units of C
- **Spray R** (per barrel) contains:
  1 unit of A, 2 units of B, and 2 units of C

Question:

How many barrels of each type of spray (P, Q, and R) should be used to provide exactly the required chemicals to control the disease?

Solution:

Let:

- x = number of barrels of Spray P
- y = number of barrels of Spray Q
- z = number of barrels of Spray R

From the given data, we form the system of equations:

$$\begin{aligned} &(\text{Chemical A}): && x + 2y + z = 8 \\ &(\text{Chemical B}): && 2x + 3y + 2z = 14 \\ &(\text{Chemical C}): && 3x + 2y + 2z = 13 \end{aligned}$$

Step 1: Matrix Form

Let:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 3 & 2 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ 14 \\ 13 \end{bmatrix}$$

The system is $AX = B$.
We solve for $X = A^{-1}B$.

Final Answer:

To control the crop disease using exactly the required amounts of chemicals:

- Use **1 barrel of Spray P**
- Use **2 barrels of Spray Q**
- Use **3 barrels of Spray R**

$$\boxed{x = 1, \quad y = 2, \quad z = 3}$$

Exercise Set:

Problem 1

Three types of food contain different units of nutrients.

- Food A: 3 units of protein, 2 units of fat, and 4 units of carbs
- Food B: 5 units of protein, 3 units of fat, and 2 units of carbs
- Food C: 4 units of protein, 4 units of fat, and 3 units of carbs

A person needs 47 units of protein, 35 units of fat, and 39 units of carbs.
Find how many units of each food should be consumed to meet the exact nutritional needs.

Problem 2

A, B, and C invested in a business by purchasing shares priced at x, y, and z respectively.

- A bought 2 shares of x, 3 of y, and 5 of z for a total of Tk. 560
- B bought 3 of x, 4 of y, and 2 of z for Tk. 470
- C bought 4 of x, 2 of y, and 3 of z for Tk. 530

Determine the price of each share using matrix method.

Problem 3

Problem 4

A chemistry lab requires exactly 24g of substance A, 36g of B, and 30g of C.
Three chemical solutions contain:

Solution	A	B	C
S1	2	4	3
S2	3	2	5
S3	1	3	2

Find how many units of each solution to mix.

Problem 5

A mobile phone manufacturer uses components X, Y, and Z.

- Phone Model A uses: 1X, 2Y, 3Z
- Phone Model B uses: 2X, 3Y, 1Z
- Phone Model C uses: 3X, 2Y, 2Z

To produce an order of phones, the manufacturer needs 36 units of X, 38 of Y, and 33 of Z.
How many of each model should be produced?

Problem 6

A juice factory blends three juices to get a new mix:

- Juice 1: 2L apple, 1L orange, 1L mango
- Juice 2: 1L apple, 2L orange, 2L mango
- Juice 3: 1L apple, 1L orange, 3L mango

The final blend requires 10L apple, 10L orange, and 16L mango.
Find how many liters of each juice to mix.

Problem 7

A mechanic uses three types of oils A, B, and C.

- Oil P: 2A, 1B, 1C
- Oil Q: 1A, 3B, 2C
- Oil R: 1A, 2B, 2C

If the requirement is 10A, 14B, and 13C units,
find the number of units of P, Q, and R to mix.

Problem 8

A carpenter makes chairs using three types of wood: W1, W2, and W3.

- Chair A uses: 2W1, 1W2, 1W3
- Chair B uses: 1W1, 2W2, 1W3
- Chair C uses: 1W1, 1W2, 2W3

To make a batch, he needs 9 units of W1, 8 of W2, and 9 of W3.
How many of each chair should he make?

Problem 9

A beverage contains 3 nutrients: calcium, iron, and vitamin D.

- Brand X: 3mg, 2mg, 4mg respectively
- Brand Y: 2mg, 3mg, 3mg
- Brand Z: 4mg, 1mg, 5mg

A diet plan requires 23mg calcium, 20mg iron, and 36mg vitamin D.
How many units of each brand should be included?

Problem 10

Three printers P, Q, and R use different amounts of toner.

- P: 3A, 2B, 1C
- Q: 2A, 3B, 2C
- R: 1A, 2B, 3C

To complete a print job, 18A, 20B, and 20C units of toner are needed.

Determine how many tasks each printer should handle.

Engineering Applications

Network Analysis

Networks composed of branches and junctions are used as models in many diverse fields such as economics, traffic analysis, and electrical engineering.

In such models it is assumed that the total flow into a junction is equal to the total flow out of the junction. For example, because the junction shown in Figure 1.9 has 25 units flowing into it, there must be 25 units flowing out of it. This is represented by the linear equation

$$x_1 + x_2 = 25.$$

Because each junction in a network gives rise to a linear equation, you can analyze the flow through a network composed of several junctions by solving a system of linear equations. This procedure is illustrated in Example 5.

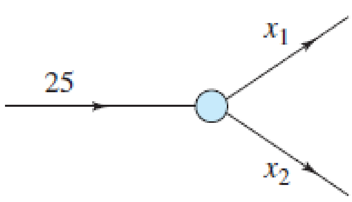

Figure 1.9

Problem-1:

Analysis of a Network

Set up a system of linear equations to represent the network shown in Figure 1.10, and solve the system.

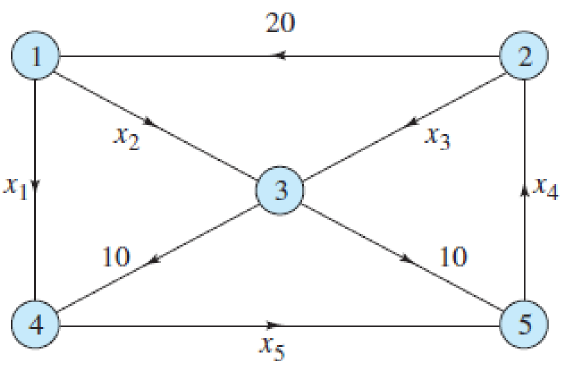

Figure 1.10

SOLUTION Each of the network's five junctions gives rise to a linear equation, as shown below.

$$
\begin{array}{rcll}
x_1 + x_2 \qquad\qquad\qquad & = & 20 & \text{Junction 1} \\
x_3 - x_4 \qquad & = & -20 & \text{Junction 2} \\
x_2 + x_3 \qquad\qquad & = & 20 & \text{Junction 3} \\
x_1 \qquad\qquad\qquad - x_5 & = & -10 & \text{Junction 4} \\
-x_4 + x_5 & = & -10 & \text{Junction 5}
\end{array}
$$

Solution by Gaussian elimination

Convert the augmented matrix into the row echelon form:

$$
\left(\begin{array}{ccccc|c}
1 & 1 & 0 & 0 & 0 & 20 \\
0 & 0 & 1 & -1 & 0 & -20 \\
0 & 1 & 1 & 0 & 0 & 20 \\
1 & 0 & 0 & 0 & -1 & -10 \\
0 & 0 & 0 & -1 & 1 & -10
\end{array}\right) \times(-1)
$$

$$
\underset{R_4 - 1 \cdot R_1 \to R_4}{\sim}
\left(\begin{array}{ccccc|c}
1 & 1 & 0 & 0 & 0 & 20 \\
0 & 0 & 1 & -1 & 0 & -20 \\
0 & 1 & 1 & 0 & 0 & 20 \\
0 & -1 & 0 & 0 & -1 & -30 \\
0 & 0 & 0 & -1 & 1 & -10
\end{array}\right)
\underset{R_3 \leftrightarrow R_2}{\sim}
\left(\begin{array}{ccccc|c}
1 & 1 & 0 & 0 & 0 & 20 \\
0 & 1 & 1 & 0 & 0 & 20 \\
0 & 0 & 1 & -1 & 0 & -20 \\
0 & -1 & 0 & 0 & -1 & -30 \\
0 & 0 & 0 & -1 & 1 & -10
\end{array}\right) \times(1)
\underset{R_4 - (-1) \cdot R_2 \to R_4}{\sim}
$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & 1 & 1 & 0 & 0 & 20 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 0 & 1 & 0 & -1 & -10 \\ 0 & 0 & 0 & -1 & 1 & -10 \end{array}\right) \times(-1) \; R_4-1\cdot R_3 \to R_4 \; \sim \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & 1 & 1 & 0 & 0 & 20 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 0 & 0 & 1 & -1 & 10 \\ 0 & 0 & 0 & -1 & 1 & -10 \end{array}\right) \times(1) \; R_5-(-1)\cdot R_4 \to R_5 \; \sim \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & 1 & 1 & 0 & 0 & 20 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 0 & 0 & 1 & -1 & 10 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\begin{cases} x_1 + x_2 = 20 \\ x_2 + x_3 = 20 \\ x_3 - x_4 = -20 \\ x_4 - x_5 = 10 \end{cases} \quad (1)$$

- Find the variable x_4 from the equation 4 of the system (1):
  $x_4=10+x_5$
- Find the variable x_3 from the equation 3 of the system (1):
  $x_3=-20+x_4=-20+(10+x_5)=-10+x_5$
- Find the variable x_2 from the equation 2 of the system (1):
  $x_2=20-x_3=20-(-10+x_5)=30-x_5$
- Find the variable x_1 from the equation 1 of the system (1):
  $x_1=20-x_2=20-(30-x_5)=-10+x_5$

Answer:

$x_1=-10+x_5$

$x_2=30-x_5$

$x_3=-10+x_5$

$x_4=10+x_5$

$x_5=x_5$

General Solution: $X=\begin{pmatrix} -10+x_5 \\ 30-x_5 \\ -10+x_5 \\ 10+x_5 \\ x_5 \end{pmatrix}$

Home work:

Network Analysis

21. Water is flowing through a network of pipes (in thousands of cubic meters per hour), as shown in Figure 1.15.

(a) Solve this system for the water flow represented by x_i, $i = 1, 2, \ldots, 7$.
(b) Find the water flow when $x_6 = x_7 = 0$.
(c) Find the water flow when $x_5 = 1000$ and $x_6 = 0$.

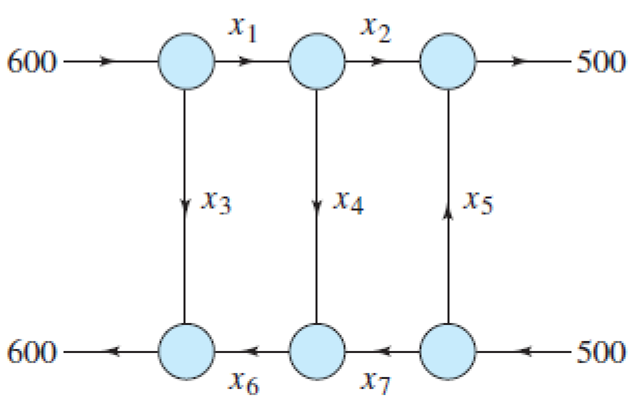

Figure 1.15

22. The flow of traffic (in vehicles per hour) through a network of streets is shown in Figure 1.16.

(a) Solve this system for x_i, $i = 1, 2, \ldots, 5$.
(b) Find the traffic flow when $x_2 = 200$ and $x_3 = 50$.
(c) Find the traffic flow when $x_2 = 150$ and $x_3 = 0$.

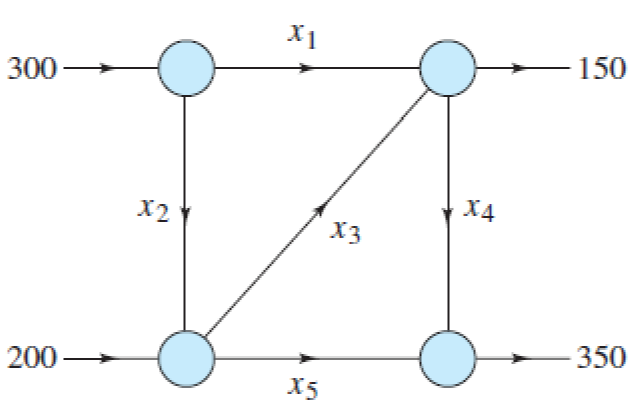

Figure 1.16

23. The flow of traffic (in vehicles per hour) through a network of streets is shown in Figure 1.17.

(a) Solve this system for x_i, $i = 1, 2, 3, 4$.
(b) Find the traffic flow when $x_4 = 0$.
(c) Find the traffic flow when $x_4 = 100$.

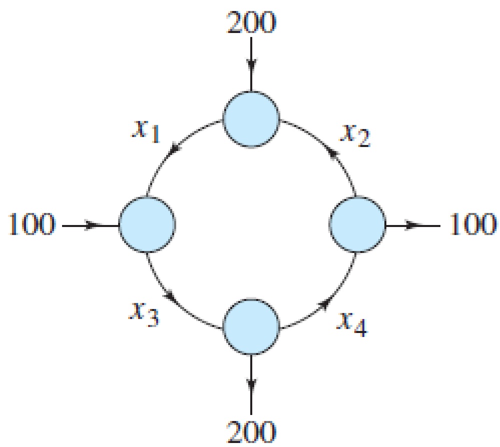

Figure 1.17

24. The flow of traffic (in vehicles per hour) through a network of streets is shown in Figure 1.18.

(a) Solve this system for x_i, $i = 1, 2, \ldots, 5$.
(b) Find the traffic flow when $x_3 = 0$ and $x_5 = 100$.
(c) Find the traffic flow when $x_3 = x_5 = 100$.

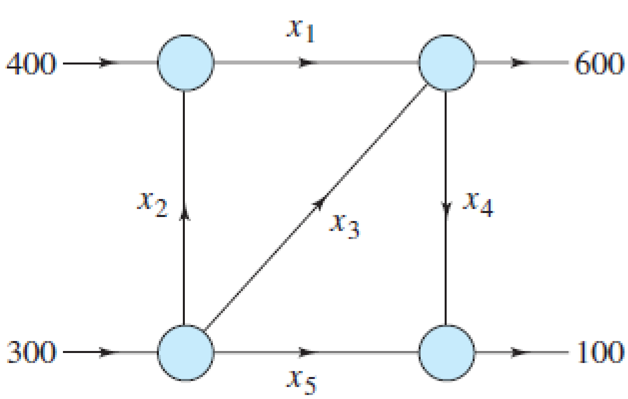

Figure 1.18

An electrical network is another type of network where analysis is commonly applied. An analysis of such a system uses two properties of electrical networks known as **Kirchhoff's Laws.**

1. All the current flowing into a junction must flow out of it.
2. The sum of the products IR (I is current and R is resistance) around a closed path is equal to the total voltage in the path.

In an electrical network, current is measured in amps, resistance in ohms, and the product of current and resistance in volts. Batteries are represented by the symbol . The larger vertical bar denotes where the current flows out of the terminal. Resistance is denoted by the symbol . The direction of the current is indicated by an arrow in the branch.

REMARK: A *closed* path is a sequence of branches such that the beginning point of the first branch coincides with the end point of the last branch.

EXAMPLE 6 Analysis of an Electrical Network

Determine the currents I_1, I_2, and I_3 for the electrical network shown in Figure 1.13.

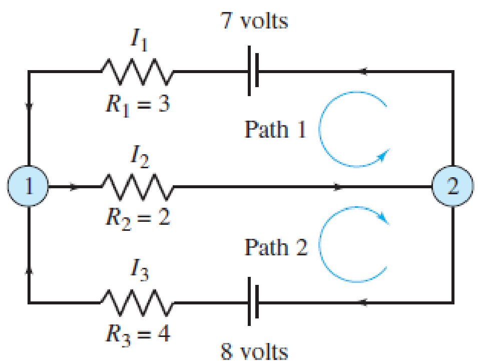

Figure 1.13

SOLUTION Applying Kirchhoff's first law to either junction produces

$$I_1 + I_3 = I_2 \qquad \text{Junction 1 or Junction 2}$$

and applying Kirchhoff's second law to the two paths produces

$$R_1I_1 + R_2I_2 = 3I_1 + 2I_2 = 7 \qquad \text{Path 1}$$
$$R_2I_2 + R_3I_3 = 2I_2 + 4I_3 = 8. \qquad \text{Path 2}$$

So, you have the following system of three linear equations in the variables I_1, I_2, and I_3.

$$\begin{aligned} I_1 - I_2 + I_3 &= 0 \\ 3I_1 + 2I_2 \qquad &= 7 \\ 2I_2 + 4I_3 &= 8 \end{aligned}$$

Applying Gauss-Jordan elimination to the augmented matrix

$$\begin{bmatrix} 1 & -1 & 1 & 0 \\ 3 & 2 & 0 & 7 \\ 0 & 2 & 4 & 8 \end{bmatrix}$$

produces the reduced row-echelon form

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

which means $I_1 = 1$ amp, $I_2 = 2$ amps, and $I_3 = 1$ amp.

Solution by Cramer's rule:

$$\Delta = \begin{vmatrix} 1 & -1 & 1 \\ 3 & 2 & 0 \\ 0 & 2 & 4 \end{vmatrix} = 26$$

$$\Delta_1 = \begin{vmatrix} 0 & -1 & 1 \\ 7 & 2 & 0 \\ 8 & 2 & 4 \end{vmatrix} = 26;$$

$$\Delta_2 = \begin{vmatrix} 1 & 0 & 1 \\ 3 & 7 & 0 \\ 0 & 8 & 4 \end{vmatrix} = 52;$$

$$\Delta_3 = \begin{vmatrix} 1 & -1 & 0 \\ 3 & 2 & 7 \\ 0 & 2 & 8 \end{vmatrix} = 26;$$

$$x_1=\Delta_1/\Delta=\frac{26}{26}=1$$
$$x_2=\Delta_2/\Delta=\frac{52}{26}=2$$
$$x_3=\Delta_3/\Delta=\frac{26}{26}=1$$

Answer:

$$x_1=1$$
$$x_2=2$$
$$x_3=1$$

Home Work:

25. Determine the currents I_1, I_2, and I_3 for the electrical network shown in Figure 1.19.

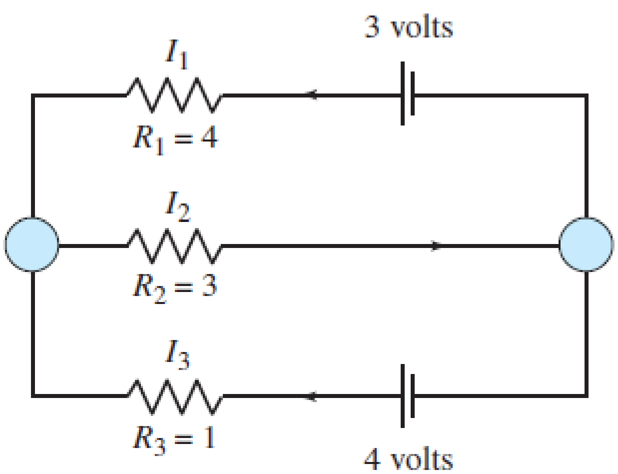

Figure 1.19

26. Determine the currents I_1, I_2, and I_3 for the electrical network shown in Figure 1.20.

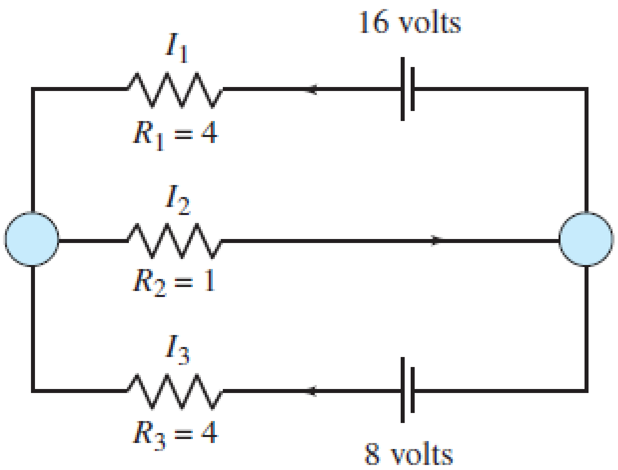

Figure 1.20

27. (a) Determine the currents I_1, I_2, and I_3 for the electrical network shown in Figure 1.21.

(b) How is the result affected when A is changed to 2 volts and B is changed to 6 volts?

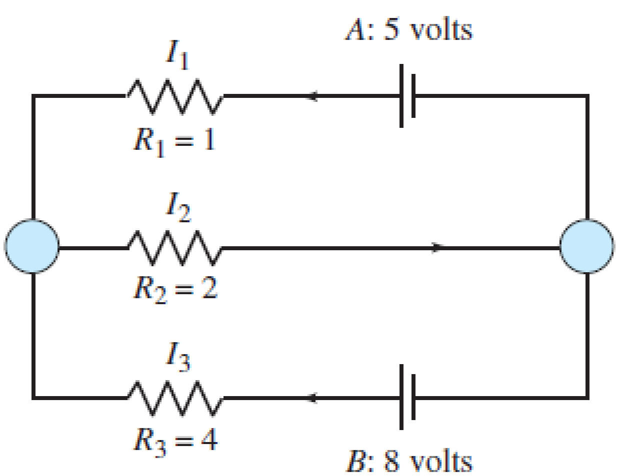

Figure 1.21